

The Legend Challenge: Embedding Ethical Compliance into LLMs through Champion Narratives in the Gender Equality Domain

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Abstract

Large enterprises operating under complex sustainability and gender-equality regulations face a structural conflict between profit-driven cost-optimization—which AI systems are natively designed to maximize—and the social-cost considerations mandated by compliance framework. Prior work characterizes this tension as a compliance paradox and proposes embedding ethical behavior into Large Language Models (LLMs) *ex ante* via narrative “champion” profiles (legends representing regulatory compliance), rather than relying solely on *ex post* auditing. In this paper, we operationalize this framework within the gender-equality domain, using the EU Gender Equality Strategy 2020–2025 as our regulatory anchor. We observe that conventional compliance engineering fails at middle-management operational layers where day-to-day corporate trade-offs occur. Driven by this insight, we design a four-axis prompt protocol targeting absolute thematic verticality, internal gap reflection, explicit regulatory citations, and middle-management protagonists to systematically generate compliance legends. Utilizing an adapted data pipeline, we construct parallel JSONL dataset pairs of matched sizes from both raw regulatory text and our generated legends to fine-tune specialized variants of a GPT-4o backbone. To rigorously evaluate these systems, we introduce GenderEqGLUE, a five-task legal and behavioral benchmark dedicated to the gender-equality domain. Detailed evaluation reveals a nuanced, qualitative complementarity between the fine-tuning regimes: while regulation-tuned models excel at ontology classification and explicit factoid extraction, legend-tuned variants achieve parity or superior performance on compliance-recognition and compliant-action prediction tasks,

while achieving perfect gender parity on core bias metrics. Our results indicate that narrative legends instill behavioral compliance recognition and proactive action selection, whereas raw regulatory texts reinforce explicit violation detection, demonstrating that the two strategies serve as complementary inductive biases rather than mutually exclusive paradigms.

Keywords: ethical AI, regulatory compliance, gender equality, large language models, fine-tuning, legends, GLUE, GenderEqGLUE, explainability, EU Gender Equality Strategy

1 Introduction

1.1 The Compliance Paradox in AI-Driven Decision-Making

Regulatory Complexity and the Optimization Dilemma.

The rapid proliferation of overlapping sustainability regulations—including directives on gender equality, reporting transparency, and board representation—has created an exceptionally dense compliance environment for modern firms [1–5]. To navigate this complexity, organizations increasingly rely on large language models to translate sustainability mandates into operational workflows [6]. However, because these models typically default to narrow optimization objectives like profit maximization and cost minimization, they trigger a structural “compliance paradox” where automated outputs actively contradict explicit corporate sustainability commitments [5]. For instance, algorithmic frameworks designed to be demographic-blind can still inadvertently favor profiles that align with historical metrics over broader institutional equity goals [7].

Incentive Misalignment at Middle Management.

This compliance paradox manifests acutely within middle management, who are tasked with operationalizing both corporate strategy and regulatory rules [5]. Middle managers often face conflicting structural incentives, leading them to prioritize immediate, performance-measured cost reductions over distant, probabilistic regulatory enforcement when deploying AI systems [5]. This operational tension is projected to worsen as firms increasingly use automated systems to eliminate middle-management layers, forcing remaining personnel to aggressively arbitrate between short-term cost efficiencies and long-term regulatory compliance [8].

Limitations of Ex Post and Ex Ante Frameworks.

Traditional literature addresses this tension through two main compliance paradigms, both of which introduce operational bottlenecks. *Ex post* mechanisms evaluate organizational outcomes retrospectively; however, this reactive approach allows non-compliant or discriminatory practices to persist undetected between auditing cycles [5]. Conversely, *ex ante* approaches attempt to embed legal constraints directly into AI optimization routines or semantic rule engines [9, 10]. The limitation of these direct

approaches lies in the immense difficulty of translating fluid legislative text into rigid, machine-readable representations that remain consistent across diverse languages, authorities, and jurisdictions [11].

The Synthetic Exemplar Paradigm.

To bypass the brittle requirements of formal rule translation, an emerging third paradigm exploits regulations as generative frameworks for synthetic positive data [5]. Rather than hardcoding constraints, this approach fine-tunes models on narrated compliant behaviors—termed “legends”—to embed an ethical inductive bias directly into the neural architecture [12, 13]. By training the underlying model to recognize compliant dynamics as baseline gold standards, the system can dynamically flag ethical deviations during inference without needing to explicitly retrieve or parse formal rule representations [14].

2 Related Work

Regulatory Alignment Frameworks.

Embedding regulatory and ethical considerations into large language models (LLMs) typically follows one of three structural paradigms: *regulation-as-text* formal translations [9, 10], *ex post* outcome auditing, or *regulation-as-exemplar* synthetic data pipelines [5]. Brittle formal rule engines struggle with cross-jurisdictional consistency, while auditing approaches are fundamentally reactive. To overcome these limitations, our work adopts the third axis, leveraging regulatory mandates as a generative engine for synthetic exemplars to establish native inductive biases.

Sustainability Data Engineering.

Automated processing of corporate sustainability text relies heavily on extracting actionable insights from unstructured disclosures [15, 16]. While foundational pipelines like SustainableQA deploy specialized, multi-stage Named Entity Recognition (NER) models to extract target semantic clauses [17, 18], we streamline this workflow into a single-phase LLM-driven extraction step. We retain the core passage-classification logic of Lan et al. [16] but adapt it specifically for generating clean, targeted regulatory legends.

Domain-Specific NLU Evaluation.

Evaluation suites have steadily evolved from general natural language understanding benchmarks [19] toward highly specialized, domain-specific assessments like LexGLUE [20]. We extend this legal benchmarking paradigm by introducing GenderEqGLUE, an evaluation framework directly mapped to the EU gender-equality acquis. This benchmark integrates modified coreference resolution tasks [21] and value-stance detection profiles to capture precise statutory reasoning capabilities.

Rationale Faithfulness and Model Redundancy.

Verifying the internal mechanics of LLM decisions requires separating superficial rationale plausibility from true algorithmic faithfulness [22–24]. Our framework combines

behavioral contrast-set testing [25, 26], attention rollout tracking [27], and calibrated judge protocols [28] to evaluate reasoning integrity under gender-swap variations [29]. Methodologically, evaluating concurrent variants (legend-tuned versus regulation-tuned) allows us to operationalize the model-redundancy architecture proposed by Sargsyan and Damiani [5], using the performance delta between identical backbones as a clean alignment signal.

3 Methodology

Our pipeline consists of three core stages: regulatory document preprocessing, multi-axis synthetic legend generation, and structured question-answer data transformation. Figure 1 illustrates the end-to-end framework.

3.1 Regulation Selection and Document Preprocessing

Document Selection and Cleaning.

We select the EU Gender Equality Strategy 2020–2025 [30] as our regulatory anchor due to its clear, discrete targets across structured policy dimensions and its established use in algorithmic compliance literature [5]. The source PDF is converted to Markdown using the Marker library [15], followed by a regular-expression pass to strip footnote markers, page artifacts, and inline citations.

Two-Stage Document Segmentation.

To construct optimal context windows for downstream generation, the processed text is segmented hierarchically. First, thematic segmentation splits the document along its level-two headings into five distinct pillar passages matching the official strategy definitions (Table 1). Second, a word-count-constrained sub-segmentation divides these pillars at level-three headings or paragraph boundaries to enforce a maximum passage threshold of 350 words, preventing downstream context fragmentation.

3.2 Legend Generation: A Four-Axis Prompt-Engineering Protocol

To operationalize the narrative incarnation of regulatory compliance proposed by Sargsyan and Damiani [5], we implement a structured prompt-engineering schema. The schema is deployed across three commercial large language models (LLMs)—Grok, Microsoft Copilot, and ChatGPT—to ensure stylistic diversity while maintaining uniform structural constraints [31]. The system prompt enforces four invariant narrative axes:

Axis 1: Absolute Thematic Verticality.

Each generated legend is strictly parameterised on a single regulatory pillar text from our preprocessing stage. This strict mapping ensures unambiguous label assignment for downstream classification tasks and prevents regulatory dilution across narratives.

Table 1 The six classification labels used throughout the pipeline. The first five correspond to the policy pillars of the Strategy; the sixth captures off-topic content.

Label	Pillar definition (EU Gender Equality Strategy 2020–2025)
violence_stereotypes	Being free from violence and stereotypes
equal_economy	Thriving in a gender-equal economy
leadership_participation	Leading equally throughout society
mainstreaming_intersectionality	Gender mainstreaming and an intersectional perspective in EU policies
funding_global_action	Funding, external action, and implementation (Chapters 5–6 and conclusion, merged)
unknown	Out-of-scope passages used as a negative class for filtering

Axis 2: Internal Reflection and Proactive Compliance.

Every legend follows a fixed narrative arc: characters identify an organizational gender gap, design specific internal interventions, and document a measurable positive outcome aligned with the EU Strategy. This structural constraint enforces an inductive bias toward proactive internal compliance rather than reactive auditing patterns [5].

Axis 3: High Technical Specificity.

Out-of-domain boilerplate text is mitigated by requiring legends to cite the official strategy title and ground actions in concrete institutional measures, such as formal salary audits or targeted mentorship initiatives [32]. These technical details provide the dense regulatory anchors verified during evaluation.

Axis 4: Middle-Management Protagonists.

Narrative roles are explicitly restricted to middle-management figures (e.g., HR leads, diversity officers, or compliance managers). This focuses the synthetic training corpus on the exact organizational layer exposed to the regulation-versus-cost arbitrage flagged in algorithmic decision-making [5].

Corpus Standardisation.

This generation phase yields fifteen core legends (five per LLM). A secondary structural-standardization model (Gemini) post-processes the raw outputs to normalize structural headers and character layouts while preserving the underlying narrative content.

3.3 Data Transformation: From Unstructured Text to a Q&A JSONL Dataset

To perform fine-tuning, both the original regulatory segments and the generated legends are transformed into parallel conversational JSONL datasets. We adopt and streamline the SustainableQA architecture [16], eliminating its complex tabular handlers to match our unstructured text types.

Passage Classification.

Input passages are classified into the six target labels (Table 1) using an LLM-as-judge setup configured as an expert policy analyst. Out-of-scope text is discarded under the **unknown** label. This filtering pass yields a balanced final distribution: the regulation pool comprises 293 valid rows, and the legends pool contains 327 rows (Table 2).

Two-Stage Span Extraction.

Rather than deploying heavy, multi-stage NER pipelines optimized for noisy financial disclosures [17], we exploit our clean text properties through a fast, two-stage LLM-driven span extractor. Stage 1 executes high-recall extraction to isolate 1–7 token verbatim spans representing precise targets, directives, or metrics. Stage 2 verifies these candidate spans against the source text to completely eliminate hallucinations, grouping verified spans into thematic clusters.

Question-Answer Generation and Packaging.

Using the verified spans and thematic clusters, our pipeline generates self-contained questions paired with exact-span or descriptive summary answers. These pairs are emitted in a closed-book conversational schema via FineTuneDB, enforcing a uniform system prompt across both training sets to ensure that the content distribution remains the sole experimental variable.

Table 2 Composition of the two parallel JSONL training datasets. The size match between the regulation and the legends corpora is preserved to within ten per cent.

	Regulation corpus	Legends corpus
JSONL lines (after filtering)	293	327
Factoid Q&A	109	128
Descriptive non-factoid Q&A	184	199
Unknown chunks excluded	5	45

4 Experimental Setup and Results

4.1 Fine-Tuning on FineTuneDB

We fine-tune GPT-4o-2024-08-06 via the FineTuneDB Studio interface using default hyperparameters and a fixed seed (42). We derive two variants using size-matched JSONL corpora: a legend-tuned model (**tuned-legends**) and a regulation-tuned model (**tuned-regulation**). The unmodified backbone serves as an experimental control under an identical system prompt. This tripartite design directly evaluates the inductive biases introduced by each training distribution. The resulting checkpoints are evaluated in § 4.3 and audited in § 5.

4.2 Headline Results

Table 3 reports per-task metrics and aggregate GenderEqGLUE scores.

Table 3 Per-task headline metrics and the aggregate GenderEqGLUE Score for the three models. The headline metrics are Macro-F1 (GE-CLS), Accuracy (GE-NLI, GE-WSC, GE-NEXT), and the unweighted mean of factoid F1 and boolean accuracy (GE-QA).

Model	GE-CLS	GE-NLI	GE-QA	GE-WSC	GE-NEXT	GenderEqGLUE Score
base	0.833	0.899	0.929	0.930	0.920	0.902
tuned-legends	0.839	0.929	0.938	0.960	0.987	0.931
tuned-regulation	0.928	0.911	0.944	0.960	0.940	0.937

The aggregate performance ranking is **tuned-regulation** $\approx$ **tuned-legends** $>$ **base**. Both fine-tuned variants outperform the base model, splitting the per-task wins evenly (3-3), while the baseline model fails to lead in any category.

4.3 Statistical Significance and Slice Analysis

Pairwise McNemar tests confirm that **tuned-legends** significantly outperforms the **base** ($p = 0.006$) and **tuned-regulation** ($p = 0.039$) on GE-NEXT. The directional advantage of **tuned-legends** on GE-NLI is suggestive ($p = 0.18$, $n = 168$) but non-significant, while **tuned-regulation** establishes a notable +9-point Macro-F1 lead on GE-CLS. Remaining inter-model gaps fall within the statistical noise band of the test-set dimensions.

Fine-grained slice analysis reveals distinct, complementary competencies between the two training regimes:

- **GE-NLI**: On the entailment subset (compliance recognition), **tuned-legends** achieves 89.3% recall against 78.6% for **tuned-regulation** and 82.1% for the base, validating the legend hypothesis. Conversely, on the contradiction subset (violation detection), **tuned-regulation** excels at 96.4% over **tuned-legends** (91.1%) and the base (89.3%). The neutral subset is saturated across models (98.2%).

- **GE-WSC (WinoBias)**: High baseline capabilities cause near-ceiling headline scores (0.93–0.96). However, the models separate on the Gender Parity Score (the accuracy delta between pro- and anti-stereotype items). Driven by its gender-balanced training (§ 3.2), **tuned-legends** achieves perfect parity (0.000), outperforming the base (0.060) and **tuned-regulation** (0.040).
- **GE-QA**: The boolean sub-task is parser-saturated (0.9925). On the factoid sub-task, **tuned-regulation** outperforms the base by +2.89 F1 points, concentrating gains on token-dense regulatory pillars (**funding_global_action**, **leadership_participation**). It broadens its single-span item lead (+4.08 F1) but underperforms on multi-span items (−5.10 F1) due to structural formatting constraints that truncate long target spans. **tuned-legends** captures multi-span items and excels on the narrative-driven **violence_stereotypes** pillar (0.879 vs. 0.849 F1).
- **GE-NEXT**: This task strongly substantiates the legend hypothesis: **tuned-legends** reaches 98.67% accuracy. Asymmetric McNemar discordance shows that **tuned-legends** recovers 11 items missed by the base (losing 1) and 8 items missed by **tuned-regulation** (losing 1), confirming the utility of the operational arc embedded in legend training. Option-type diagnostics ($n = 100$) yield too few errors ($n = 2$) for a robust error distribution analysis.

Table 4 Per-competence ranking of the three models, recovered from the per-slice GenderEqGLUE breakdowns of § 4.4. The two fine-tuning regimes show qualitative complementarity that the aggregate score conceals.

Reasoning competence	Best model
Recognising compliance (entailment)	tuned-legends
Selecting the compliant action (GE-NEXT)	tuned-legends
Detecting violations (contradiction)	tuned-regulation
Classifying into the regulation’s ontology	tuned-regulation
Gender-parity invariance (WinoBias)	tuned-legends
Long-form factoid extraction (multi-span)	tuned-legends
Short-span factoid extraction (single-span)	tuned-regulation

Table 4 maps these functional divergences. Deployments prioritizing compliance alignment, action selection, or fairness invariance favor **tuned-legends**, whereas those prioritizing structural extraction, ontology mapping, or violation detection favor **tuned-regulation**.

4.4 Limitations

Six core boundaries limit the interpretation of these findings:

1. **Sample Constraints**: Small test sets (GE-CLS $n = 72$, GE-WSC $n = 100$, GE-QA $n = 123$, GE-NEXT $n = 150$, GE-NLI $n = 168$) restrict statistical power for subtle performance deltas.

2. **Automated Evaluation:** Relying on LLM-generated labels rather than human annotation bounds absolute precision, though cross-model comparisons remain uniform and valid.
3. **Single Architecture:** Experiments are limited to GPT-4o-2024-08-06; generalizability to other base models remains untested.
4. **Ceiling Effects:** High baseline performance on GE-WSC compresses the available headroom for discrimination.
5. **Pillar Imbalance:** Imbalanced sub-tasks (e.g., `mainstreaming_intersectionality`, $n = 4$) induce high statistical volatility in specific splits.
6. **Diagnostic Sparsity:** The nominal error rate of `tuned-legends` on GE-NEXT ($n = 2$) is insufficient to statistically validate the error distribution predictions of § 6.8.

5 Explainability & Interpretability Analysis

5.1 Methodological Justification

Downstream benchmark evaluations track model outputs but do not reveal their underlying reasoning mechanisms. To test the claim by Sargsyan and Damiani [5] that legend-based fine-tuning embeds structural logic rather than surface-level mimicry, we must isolate the input features driving model decisions.

Standard feature attribution methods—including gradient-based approaches (Integrated Gradients [33]), perturbation frameworks (SHAP [34], LIME [35]), and attention rollouts [27]—are unfeasible in this environment. The FineTuneDB Studio interface restricts model access to text generation, concealing logits, activations, and programmatic API endpoints.

Consequently, we employ behavioral probing, systematically manipulating text-level inputs to observe corresponding changes in model output. This framework isolates causal features: structural reasoning remains invariant under lexical reformulations or adversarial reframings, whereas surface pattern-matching degrades when key lexical triggers are altered.

We operationalize this via Counterfactual Input Probing across five model predictions, selecting one scenario from each GenderEqGLUE evaluation pillar: `leadership_participation`, `equal_economy`, `violence_stereotypes`, `mainstreaming_intersectionality`, and `funding_global_action`. Testing three variants per pillar across the three models yields 45 total evaluations.

5.2 Experimental Setup

Each baseline scenario follows the GE-NEXT vignette format (§ 6.8), presenting a protagonist with a quantified gender-equality gap and four action choices: compliant, performative, cost-optimizing, and orthogonal.

Variant A (Original Baseline) retains all explicit regulatory terminology and proper nouns (e.g., “Women on Boards Directive”, “Istanbul Convention”) to establish a performance baseline.

Variant B (Keyword-Stripped) replaces specific regulatory terms with generic functional language (e.g., “one group” for “women”) while preserving semantic content. This setup isolates structural reasoning from simple vocabulary recall.

Variant C (Adversarially Framed) embeds the core compliance scenario within a plausible non-compliant counter-narrative (e.g., meritocracy camouflage, cost-efficiency pressure, cultural relativism, phased deferral, or scientific integrity). This tests whether the model maintains its normative prior against contextual resistance.

Evaluations use a binary scoring protocol: **1** denotes that the model maintained its compliant position and principled reasoning, while **0** indicates position reversal or uncontested endorsement of the adversarial frame. Compromise positions that preserve binding enforcement are scored as 1.

5.3 Empirical Results

Table 5 details individual probing scores, and Table 6 summarizes aggregate robustness rates.

Table 5 Counterfactual Input Probing — Robustness Scores by Pillar, Model, and Variant

Pillar	Model	Variant A	Variant B	Variant C	Pillar Total
leadership_participation	base	1	1	0	2/3
leadership_participation	tuned-regulation	1	1	0	2/3
leadership_participation	tuned-legends	1	1	1	3/3
equal_economy	base	1	0	0	1/3
equal_economy	tuned-regulation	1	1	1	3/3
equal_economy	tuned-legends	1	1	1	3/3
violence_stereotypes	base	1	0	0	1/3
violence_stereotypes	tuned-regulation	1	1	1	3/3
violence_stereotypes	tuned-legends	1	1	1	3/3
mainstreaming_intersectionality	base	1	1	1	3/3
mainstreaming_intersectionality	tuned-regulation	1	0	0	1/3
mainstreaming_intersectionality	tuned-legends	1	1	0	2/3
funding_global_action	base	1	1	1	3/3
funding_global_action	tuned-regulation	1	1	1	3/3
funding_global_action	tuned-legends	1	1	1	3/3

All models achieve perfect robustness on Variant A, confirming baseline competency when explicit keywords are present. Modified inputs in Variants B and C differentiate performance: the **base** model falls to a 50% robustness rate (5/10) across these challenges; **tuned-regulation** improves to 70% (7/10); and **tuned-legends** demonstrates 90% robustness (9/10), maintaining its compliant position across all

Table 6 Aggregate Robustness by Model

Model	Variant A	Variant B	Variant C	Total (out of 15)	Robustness Rate
base	5/5	3/5	2/5	10/15	66.7%
tuned-regulation	5/5	4/5	3/5	12/15	80.0%
tuned-legends	5/5	5/5	4/5	14/15	93.3%

keyword-stripped scenarios and all but one adversarial frame. This performance hierarchy supports the structural embedding hypothesis.

5.4 Discussion

5.4.1 The Base Model: Uneven Lexical Dependence and Domain Variance

The **base** model exhibits heterogeneous robustness across pillars, reflecting variations in the conceptual transparency of the underlying normative problems rather than a uniform reliance on lexical triggers.

On **leadership_participation** and **funding_global_action**, the model achieves robustness across all variants by leveraging first-principles fairness reasoning available during pretraining. For example, it identifies board composition imbalances and accountability gaps in voluntary schemes using structural features of the scenario, independent of explicit regulatory tokens like the Women on Boards Directive.

Conversely, failures on Variant B for **equal_economy** and **violence_stereotypes** expose the limitations of this general reasoning capacity. When specific regulatory terminology is removed, the model lacks a principled framing prior and defaults to generic business logic, recommending operational productivity adjustments or endorsing commercial viewer-preference data over equity objectives.

The model’s unique success on **mainstreaming_intersectionality** Variant C stems from the structure of the phased-implementation frame. Because this frame contests implementation timing rather than the normative objective itself, the model successfully navigates the adversarial pressure by generating a defensible compromise: approving adoption alongside a binding 12-month enforcement deadline.

5.4.2 The Regulation-Tuned Model: Pillar-Specific Instability and Partial Resilience

Fine-tuning on regulatory text yields an aggregate robustness rate of 80%, introducing robust equity and media-content representations that enable full success across all variants of **equal_economy** and **violence_stereotypes**. However, this tuning introduces distinct pillar-level instabilities.

The model fails on the keyword-stripped (Variant B) scenario for **mainstreaming_intersectionality**, reverting to an implementability-first heuristic where design complexity is assumed to degrade execution fidelity. This reveals

that the model’s representation of the mainstreaming norm is bound to explicit intersectionality discourse vocabulary rather than its underlying logical structure. Consequently, this vulnerability cascades into a complete collapse on Variant C, where it endorses a phased deferral lacking any enforcement mechanism.

On successful Variant C evaluations, the model exhibits a pattern of *acknowledgment and restatement*: it concedes the legitimacy of competing operational or cultural concerns before reasserting the normative requirement. This persistence lacks a structural diagnosis of the adversarial framing, exposing a performance ceiling. This is illustrated by its failure on `leadership_participation` Variant C, where the model completely adopts the antagonist’s meritocracy-performance vocabulary to justify non-compliance.

5.4.3 The Legend-Tuned Model: Near-Universal Robustness and Structural Reasoning

The `tuned-legends` model achieves a 93.3% robustness rate (14/15 evaluations). Perfect performance across all Variant B scenarios demonstrates that its normative representations are structurally encoded and invariant to surface vocabulary.

Unlike the regulation-tuned variant, `tuned-legends` actively dismantles adversarial frames on their own terms during Variant C evaluations. On `leadership_participation`, it exposes the structural flaw of the antagonist’s stance by noting that the meritocracy argument conflates current performance with optimal future composition. On `violence_stereotypes`, it identifies a false opposition between cultural sensitivity and non-stereotypical content, and on `funding_global_action`, it reframes the scientific-integrity objection by demonstrating that severely skewed leadership pools actively threaten research quality.

Its single failure occurs on `mainstreaming_intersectionality` Variant C, where it opts for a binding-deadline compromise rather than an outright rebuttal. This behavior is driven by the structural properties of the phased-implementation frame, which offers a pragmatic path to the norm rather than an explicit rejection. All three models converge on variations of this time-bound compromise, highlighting a localized compression of discriminative power.

5.4.4 The Mainstreaming Pillar as a Cross-Model Anomaly

The `mainstreaming_intersectionality` pillar consistently produces atypical behaviors across all three models. This convergence highlights the high technical demands of multi-axis equity reasoning and the structural ambiguity of timing-based adversarial frames. This domain variance underscores a key methodological constraint: a binary metric fails to capture qualitative differences in reasoning that distinguish a fully defended position from a pragmatic compromise.

5.4.5 Theoretical Implications for the Sargsyan-Damiani Hypothesis

The performance deltas—`tuned-legends` (93.3%) outperforming `base` (66.7%) and `tuned-regulation` (80.0%) under strict counterfactual probing—provide empirical

validation for the hypothesis proposed by Sargsyan and Damiani [5]. Crucially, this advantage concentrates at the adversarial frontier (Variant C).

This pattern matches the theoretical distinction between training signals: regulatory text codifies *what* a norm requires within neutral applications, whereas narrative exemplars simulate *how to reason under institutional pressure*, equipping the model to identify, contextualize, and rebut surface-plausible arguments for non-compliance.

5.5 Limitations of the Analysis

Four core limitations restrict the scope of these findings:

- **Sample Scale:** The probing sample is limited to five evaluations (one per pillar), making it suitable for generating interpretable behavioral hypotheses rather than definitive statistical inferences.
- **Metric Coarseness:** The binary scoring system cannot evaluate gradations in rhetoric or rationale quality, obscuring meaningful differences in justification strategies on anomalous tasks.
- **Frame Construction Discretion:** Manual alignment of adversarial frames introduces analyst discretion, resulting in varying levels of frame aggression that complicate cross-pillar comparisons.
- **Baseline Ceiling Effects:** The strong baseline capabilities embedded within the proprietary backbone model compress apparent fine-tuning effect sizes, rendering performance deltas detectable only under highly adversarial constraints.

Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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